

Program Question_1

```
Implicit none
Integer,Parameter::n=6
Real(8),Parameter::TOL=1.0E-3
Integer,Parameter::M=1000
Real(8)::A(n,n),b(n),X0(n),x(n)
Real(8)::norm,sum
Integer::i,j,k

!Initialize the Matrix A

A = reshape([ &
  4.0, -1.0,  0.0, -1.0,  0.0,  0.0, &
 -1.0,  4.0, -1.0,  0.0, -1.0,  0.0, &
  0.0, -1.0,  4.0,  0.0,  0.0, -1.0, &
 -1.0,  0.0,  0.0,  4.0, -1.0,  0.0, &
  0.0,  0.0,  0.0, -1.0,  4.0, -1.0, &
  0.0,  0.0, -1.0,  0.0, -1.0,  4.0 ], [n,n])

!Initialize Vector b

b=[0,5,0,6,-2,6]
!Initalize guess
X0=0.0
x=0.0
Write(*,*)"Iter      x1      x2      x3      x4      x5      x6      Error"
Write(*,*)"_____ "

!step-1
k=1
!step-2
Do while(k<=M)
  norm=0.0
  !step_3(compute new x(i)
  Do i=1, n
    sum=0.0
    Do j=1, n
      If(j/=i)Then
        sum=sum+A(i,j)*X0(j)
      End If
    End Do
    x(i)=(b(i)-sum)/A(i,i)
  End Do
  !Step-4(Compute Infinity norm)
  Do i=1, n
    If(abs(x(i)-X0(i))>norm)Then
      norm=Abs(x(i)-X0(i))
    end if
  End Do

  !Print the current iteration
  Write(*,'(I4,7F9.5)')k,x,norm
  If(norm<TOL)Then
    Write(*,*)"The Procedure was succesful"
    Write(*,*)"Solution after",k,"Iteration"
    Do i=1, n
      Write(*,'(A,I1,A,F10.6)') "x(",i,")=" ,x(i)
    End Do
    stop
  End If
!step-5
k=k+1
!step6
X0=x
```

End do

! Step 7: Maximum number of iterations exceeded

print *, "Maximum number of iterations exceeded."

End Program